

NIKHILA PERAVALI

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EDUCATION

University of Massachusetts Amherst, Amherst, MA

Sept 2021- May 2025

Bachelor of Science in Computer Science and Mathematics Minor (Statistics with Data Science Concentration)

GPA: 3.62

RELEVANT COURSEWORK

Advanced Software Engineering, Artificial Intelligence, Scalable Web Systems, Principles of Data Science, Natural Language Processing, Database Management, Software Entrepreneurship, Human Computer Interaction, Algorithms and Data Structures

TECHNICAL CAPABILITIES

Programming & Scripting: Python, Java, C, JavaScript, TypeScript, HTML/CSS, SQL, R

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, MNE, NLTK

Web & Backend Development: React, Node.js, Express, REST APIs, MongoDB

Visualization & Tools: Tableau, Excel, Matplotlib, Git, Figma, Adobe XD

EXPERIENCE

Machine Learning Intern, AImpower.org

May 2024 – Aug 2024

- Benchmarked state-of-the-art Automatic Speech Recognition(ASR) models (OpenAI, Google, Meta, Microsoft, Otter.AI) on 50+ hours of stuttered Mandarin speech audio files to evaluate fluency bias, latency and transcription accuracy.
- Designed and deployed a custom GPT-4 translation and REST-based processing pipeline, improving annotation consistency and reducing review time by 40% for non-Mandarin speakers.
- Preprocessed 1000+ audio files and transcripts using Python, Pandas, and NLTK to standardize inputs for ASR model evaluation.
- Implemented API-driven evaluation scripts and analyzed model outputs using WER, CER, BLEU, WordNet, and GloVe, identifying 15–30% variance in accuracy based on stutter type.

Artificial Intelligence Research Assistant, Umass Amherst

Dec 2023 – Feb 2024

- Explored code generation with LLMs and synthesis tools, evaluating ChatGPT and Rosette on 50+ coding tasks across bug detection, optimization, and verification.
- Implemented Rosette-based synthesis achieving 100% functional correctness on 80% of tasks with complete specifications.
- Compared GPT's performance, noting 60% task completion with flexible but less reliable natural language prompts.
- Assessed tools across bug detection, optimization, and program verification, using Racket as a target language.
- Analyzed trade-offs in correctness, specification complexity, and language support, documenting implications for AI-assisted development workflows.

Data Analyst Intern, Arrow Sales Corporation

May 2023 – Aug 2023

- Analyzed 2+ years of sales and service data using SQL and Python, uncovering trends that reduced stockouts by 15%.
- Developed predictive models in scikit-learn to optimize maintenance schedules and cut downtime by 20%.
- Built Tableau dashboards to visualize customer behavior, service metrics, and machine failure trends.
- Automated weekly data pipeline scripts in Python, reducing reporting time by 50% and improving cross-team communication.

PROJECTS

Chatter AI: Real-Time Language Learning Assistant | *React, Tailwind CSS, Python, OpenAI API, Firebase*

- Developed an AI-powered web app that simulates a virtual language partner for real-time conversational practice, integrating OpenAI's language models and Firebase to deliver adaptive, interactive, and personalized learning sessions.
- Built the web-based interface using React and Tailwind CSS, designing responsive layouts and intuitive UI/UX for an engaging and seamless communication experience.

Seekly: Job Search Platform Development | *React, Tailwind CSS, Node.js, MongoDB*

- Developed a full-stack job search platform using React, Node.js, and MongoDB, implementing resume review, ATS scoring, recruiter messaging, and job tracking features.
- Optimized the user interface with Tailwind CSS and REST APIs, reducing query response time to under one second and increasing mobile engagement during testing.

Seizure Prediction from EEG Data | *Python, MNE, NumPy, Scikit-learn*

- Engineered a Python pipeline using MNE for EEG signal preprocessing, applying normalization and feature extraction to enhance model input quality.
- Trained and validated a Random Forest classifier in Scikit-learn, achieving over 90% accuracy while improving model robustness through cross-validation.

Leadership / Extracurricular

Rewriting the Code | Public Interest Technology Club | Women in Computer Science club | Kappa Phi Gamma Sorority, Inc.